Curved Yang-Mills gauge theories

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Abstract*

We construct a gauge theory based on principal bundles $\mathscr{P}$ equipped with a right $\mathscr{G}$-action, where $\mathscr{G}$ is a Lie group bundle instead of a Lie group. Due to the fact that a $\mathscr{G}$-action acts fibre by fibre, pushforwards of tangent vectors via a right-translation act now only on the vertical structure of $\mathscr{P}$. Thus, we generalize pushforwards using a connection on $\mathscr{G}$ which will modify the pushforward. A horizontal distribution on $\mathscr{P}$ invariant under such a modified pushforward will provide a proper notion of Ehresmann connection. For achieving gauge invariance we impose conditions on the connection 1-form μ on $\mathscr{G}$: μ has to be a multiplicative form, *i.e.* closed w.r.t. a certain simplicial differential δ on $\mathscr{G}$, and the curvature R_μ of μ has to be δ-exact with primitive ζ; μ will be the generalization of the Maurer-Cartan form of the classical gauge theory, while the δ-exactness of R_μ will generalize the role of the Maurer-Cartan equation. This introduces the notion of multiplicative Yang-Mills connections, a connection which helped classifying singular foliations and symmetry breaking. For allowing curved connections on $\mathscr{G}$ in the dynamical theory we will need to generalize the typical definition of the curvature/field strength F on $\mathscr{P}$ by adding ζ to F.

Several examples for a gauge theory with a curved μ will be provided, including the inner group bundle of the Hopf fibration $\mathbb{S}^7 \to \mathbb{S}^4$, and a classification for gauge theories with structural simple group bundles will be provided, including a classification for whether these theories admit a classical description.

Last but not least, this paper aims to accessible for beginners which is why it only assumes knowledge of classical gauge theory.

*Abbreviations used in this paper: **LGB(s)** for Lie group bundle(s), **LAB(s)** for Lie algebra bundle(s).

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1. Introduction and summary

2. Lie group bundles with connection

We will construct a gauge theory on a principal bundle equipped with an LGB-action. Thus let us first study LGBs; since we will also generalise the Maurer-Cartan form, we will study connections on LGBs with a suitable curvature equation.

That is, we will consider LGBs $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ over a smooth manifold M and we will denote LGB actions on a manifold N along a map $\Phi\colon N \to M$ as in the following:

Definition 2.1: Lie group bundle actions

for example [1, §1.6, special case of Def. 1.6.1, page 34]

Let M, N be smooth manifolds, $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ an LGB over M, and $\Phi\colon N \to M$ a smooth map. Then a **right-action of $\mathscr{G}$ on** N is a smooth map

$$\begin{aligned} N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G} &\to N, \\ (p, g) &\mapsto p \cdot g, \end{aligned}$$

satisfying the following properties:

$$\begin{aligned} \Phi(p \cdot g) &= \pi(g), \\ (p \cdot g) \cdot h &= p \cdot (gh), \\ p \cdot \mathbb{1}_{\Phi(p)} &= p \end{aligned}$$

for all $p \in N$ and $g, h \in \mathscr{G}_{\Phi(p)}$, where $\mathbb{1}$ is the neutral section of $\mathscr{G}$. Similarly one defines left actions.

In this work we will need to apply the tangent map to an LGB action, and for doing so the following proposition will be useful; for this remember that it is a standard exercise to show that the tangent bundle of the pullback $N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}$ is given by just applying the tangent functor T on all components, that is,

$$\mathrm{T}\big(N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}\big) = \mathrm{T}N \,{}_{\mathrm{T}\Phi}\!\times_{\mathrm{T}\pi_{\mathscr{G}}} \mathrm{T}\mathscr{G} \ .$$

Proposition 2.2: Differential of smooth LGB actions

Let M and N be smooth manifolds, $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ an LGB over M, $\Phi\colon N \to M$ a smooth map, and assume we have a right $\mathscr{G}$-action on N, $\Psi\colon N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G} \to N$. Then we have

$$\mathrm{T}_{(p,g)}\Psi(X, Y) = \mathrm{T}_p r_\sigma(X) + \overline{(\mu_{\mathscr{G}})_g\big(Y - \mathrm{D}_{e_x} R_\sigma\big(\mathrm{D}_x \mathbb{1}(\omega)\big)\big)}\Big|_{p \cdot g}$$

for all $(p, g) \in N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}$ and $(X, Y) \in \mathrm{T}N \,{}_{\mathrm{T}\Phi}\!\times_{\mathrm{T}\pi_{\mathscr{G}}} \mathrm{T}\mathscr{G}$, where $x := \Phi(p) = \pi_{\mathscr{G}}(g)$, σ is any (local) section of $\mathscr{G}$ with $\sigma_x = g$, $\mathbb{1}$ is the identity section of $\mathscr{G}$, and ω is an element of $\mathrm{T}_x M$ given by

$$\omega := \mathrm{T}_p\Phi(X) = \mathrm{T}_g\pi_{\mathscr{G}}(Y) \ .$$

We can also write

$$\mathrm{T}_{(p,g)}\Psi(X,Y) = \mathrm{T}_p r_\sigma(X) + \overline{(\mu_{\mathscr{G}})_g\big(Y - \mathrm{T}_x\sigma(\omega)\big)}\Big|_{p\cdot g}.$$

If Φ is a surjective submersion, then we can also write

$$\mathrm{T}_{(p,g)}\Psi(X,Y) = \mathrm{T}_p r_\sigma(X) + \mathrm{T}_g\Psi_\tau(Y) - \mathrm{T}_{e_x}(\Psi_\tau \circ R_\sigma)\big(\mathrm{T}_x\mathbb{1}(\omega)\big)$$

and

$$\mathrm{T}_{(p,g)}\Psi(X,Y) = \mathrm{T}_p r_g\Big(X - \mathrm{T}_{\mathbb{1}_x}\Psi_\tau\big(\mathrm{T}_x\mathbb{1}(\omega)\big)\Big) + \mathrm{T}_g\Psi_p\Big(Y - \mathrm{D}_{\mathbb{1}_x}R_\sigma\big(\mathrm{T}_x\mathbb{1}(\omega)\big)\Big) + \mathrm{T}_{\mathbb{1}_x}(\Phi_\tau \circ R_\sigma)\big(\mathrm{T}_x\mathbb{1}(\omega)\big),$$

where τ is any (local) section of Φ with $\tau_x = p$, and Ψ_τ is the orbit map through τ.

2.1. **Ehresmann connections ...**

2.1.1. *... as parallel transport.* In the following we will initially define connections via their corresponding parallel transport, for which we will assume that they are complete. That is, for a curve $\gamma\colon I \to M$ (I an open interval of $\mathbb{R}$ containing 0) we denote a parallel transport on fibre bundles $\pi_{\mathscr{F}}\colon \mathscr{F} \to M$ by

$$\begin{aligned}\mathrm{PT}_\gamma^{\mathscr{F}}\colon \mathscr{F}_{\gamma(0)} &\to \Gamma(\gamma^*\mathscr{F}),\\ p &\mapsto \mathrm{PT}_\gamma^{\mathscr{F}}(p),\end{aligned}$$

similarly for the parallel transport on $\mathscr{G}$. We expect a sort of equivariance for the parallel transport of $p \cdot g$:

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